

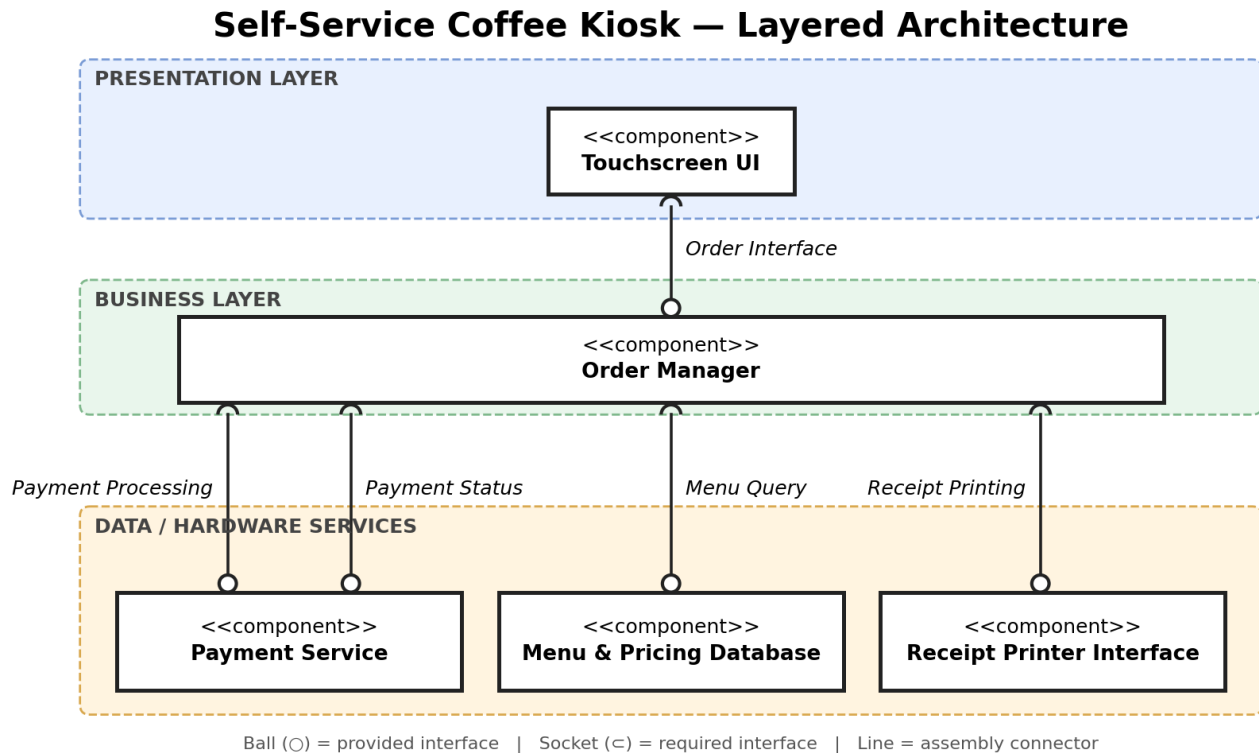
LAB 3 – COMPONENT MODELLING & ARCHITECTURE

Self-Service Coffee Kiosk System

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Component Diagram

Figure 1. Self-Service Coffee Kiosk – Layered Architecture UML Component Diagram



Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice

Layered Architecture organizes the kiosk into Presentation, Business, and Data/Hardware layers. The Touchscreen UI is in the Presentation layer; the Order Manager and Payment Service are in the Business layer; and the Menu & Pricing Database and Receipt Printer Interface support the Data/Hardware layer.

Reason 1

It provides clear separation of concerns. Customer interaction, order management, and credit-card payment processing are handled by separate components, making the system easier to test, maintain, and modify.

Reason 2

The kiosk must connect to receipt-printer hardware and store menu/pricing information. Separating these services from business logic allows the printer or stored data implementation to change without changing the core Order Manager.

Security Advantage

The Touchscreen UI does not directly access payment processing or stored data. Controlled interfaces create a boundary around the Payment Service and reduce unnecessary direct access to payment functionality.

Performance Benefit

The simple layered interaction path keeps touchscreen processing focused while the Order Manager coordinates menu lookup, payment, and receipt generation, supporting responsive interactions in a busy café.

Submission Package

This final report contains the required written architecture selection/justification and the final UML component diagram. The editable draw.io source is included separately as 480.drawio.

The diagram contains five components and the required component interactions, including Order Interface, Payment Processing, Payment Status, Menu Query, and Receipt Printing.